

# Species-Aware Review: De novo Riboflavin Biosynthesis and Flavin Cofactor Activation in *Pseudomonas putida* KT2440

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**Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00740 "Riboflavin metabolism"; module area cofactors\_vitamins\_redox **Module scope reviewed:** de novo riboflavin ring synthesis (GTP + ribulose-5-P → riboflavin) and flavin cofactor activation (riboflavin → FMN → FAD)

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## 1. Executive summary

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The de novo riboflavin biosynthesis + FMN/FAD activation module is **fully satisfiable** in *P. putida* KT2440. Every expected catalytic step maps to at least one candidate gene, and the core biosynthetic genes are organized in a recognizable chromosomal cluster (PP\_0514–PP\_0517) with dispersed additional copies. Confidence is high on architectural/homology grounds; however, **almost all of the evidence is homology-based** (UniProt/InterPro), not direct biochemistry in KT2440 itself.

The 15-gene candidate list is a mixture of (a) **7 genuine pathway enzymes**, (b) **2–3 real paralogs** that create redundancy but also naming/EC ambiguity, and (c) **~6 flavin-associated genes that are KEGG-map co-occurrence artifacts** (over-propagated into ppu00740). The most important curation actions are: exclude the peripheral flavin-utilization genes from the biosynthesis module; reclassify the two "ribAB" genes (PP\_0516, PP\_3813) as **RibB/RibBX** — active-site analysis shows their C-terminal GTP-cyclohydrolase-II-fold domain has lost the catalytic zinc-cysteine cluster and is catalytically dead, so EC 3.5.4.25 and the "ribAB" symbol should be withheld and GTP-CHII treated as a single-copy step on RibA (PP\_0522); and treat the two riboflavin-synthase paralogs as genuine duplicates rather than errors.

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## 2. Target-organism pathway definition

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**Included process:** conversion of GTP and D-ribulose 5-phosphate into the riboflavin isoalloxazine ring, then phosphorylation to FMN and adenylylation to FAD. Two converging branches (GTP-derived pyrimidine branch via GTP cyclohydrolase II + RibD; ribulose-5-P branch via DHBP synthase) meet at lumazine synthase (RibH), followed by riboflavin synthase (RibE), then the bifunctional RibF (riboflavin kinase + FMN adenylyltransferase).

**Neighboring maps/processes to keep separate (do NOT fold into this module):** - **Flavin-dependent monooxygenase electron supply** — NAD(P)H:FMN reductases (SsuE, MsuE) that feed reduced FMN to sulfonate monooxygenases (sulfur assimilation / *ssu*, *msu* systems). - **Ubiquinone/aromatic decarboxylation** — UbiX prenylated-FMN (prFMN) synthesis for UbiD-type decarboxylases (KEGG ppu00627); prFMN is a *specialized flavin derivative*, not FMN/FAD cofactor activation. - **Cobalamin (B12) biosynthesis** — BluB (5,6-dimethylbenzimidazole synthase) *consumes* reduced FMN; belongs to the B12 lower-ligand pathway. - **Nucleotide/Nudix hydrolysis** (NudF) and **haloacid dehalogenation** (Had) — not flavin metabolism at all.

**Alternate names / database definitions:** KEGG ppu00740 "Riboflavin metabolism" is broader than the biosynthesis-only module and intentionally sweeps in flavin-consuming/producing enzymes. MetaCyc/BioCyc split this into "flavin biosynthesis" + "FAD biosynthesis" sub-pathways. Gene-symbol nomenclature is inconsistent across databases: *ribC* and *ribE* both denote riboflavin synthase here, whereas in *E. coli* "ribC" historically denotes the RibF/FAD-synthetase — a known source of curation confusion.

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### 3. Expected step model and satisfiability

| # | Step (activity, EC)                  | Expected family | KT2440 gene(s)                                                                                 | Status                          |
|---|--------------------------------------|-----------------|------------------------------------------------------------------------------------------------|---------------------------------|
| 1 | GTP cyclohydrolase II (EC 3.5.4.25)  | RibA / RibBA    | <b>PP_0522 (ribA, reviewed)</b> only; PP_0516/PP_3813 C-domains are catalytically dead (RibBX) | <b>covered</b><br>(single copy) |
| 2 | DRAP deaminase (EC 3.5.4.26)         | RibD N-domain   | <b>PP_0514 (ribD)</b>                                                                          | <b>covered</b>                  |
| 3 | HTP reductase (EC 1.1.1.193)         | RibD C-domain   | <b>PP_0514 (ribD)</b>                                                                          | <b>covered</b>                  |
| 4 | DHBP synthase (EC 4.1.99.12)         | RibB / RibBA    | <b>PP_0530 (ribB, reviewed)</b> ; DHBP domain of PP_0516, PP_3813                              | <b>covered</b>                  |
| 5 | Lumazine synthase (EC 2.5.1.78)      | RibH            | <b>PP_0517 (ribH, reviewed)</b>                                                                | <b>covered</b>                  |
| 6 | Riboflavin synthase (EC 2.5.1.9)     | RibE / RibC     | <b>PP_0515 (ribE, operonic)</b> ; paralog PP_2916 (ribC)                                       | <b>covered</b>                  |
| 7 | Riboflavin kinase (EC 2.7.1.26)      | RibF N-term     | <b>PP_0602 (ribF)</b>                                                                          | <b>covered</b>                  |
| 8 | FMN adenylyltransferase (EC 2.7.7.2) | RibF C-term     | <b>PP_0602 (ribF)</b>                                                                          | <b>covered</b>                  |

**All eight catalytic activities are encoded.** The bifunctional RibF (PP\_0602, 316 aa) performs both activation steps 7+8, as is typical for bacteria (

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RibD (PP\_0514, 378 aa) is genuinely bifunctional (deaminase + reductase domains). No step is a gap; no step is "not expected."

## 4. Candidate genes and evidence

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### High-confidence pathway enzymes (7)

- **PP\_0514 ribD (Q88QH9, 378 aa, TrEMBL).** Bifunctional deaminase/reductase (EC 3.5.4.26 + 1.1.1.193). Operonic. Covers steps 2–3. Evidence: homology + domain architecture.
- **PP\_0515 ribE (Q88QH8, 221 aa, TrEMBL).** Riboflavin synthase (EC 2.5.1.9); InterPro: paired Lumazine-binding + riboflavin-synthase  $\beta$ -barrel domains. Operonic → canonical copy. Covers step 6.
- **PP\_0517 ribH (Q88QH6, 158 aa, reviewed).** Lumazine synthase (EC 2.5.1.78). Covers step 5. Highest-quality annotation (Swiss-Prot).
- **PP\_0522 ribA (Q88QH1, 205 aa, reviewed).** Monofunctional GTP cyclohydrolase II (EC 3.5.4.25). Covers step 1. Swiss-Prot.
- **PP\_0530 ribB (Q88QG4, 216 aa, reviewed).** Monofunctional DHBP synthase (EC 4.1.99.12). Covers step 4. Swiss-Prot.
- **PP\_0602 ribF (Q88Q93, 316 aa, TrEMBL).** Bifunctional riboflavin kinase / FMN adenylyltransferase (EC 2.7.1.26 + 2.7.7.2). Covers steps 7–8.
- *Note:* the three most central biosynthetic genes (ribD, ribE, ribF) are **unreviewed TrEMBL**, whereas peripheral FMN reductases are reviewed — an inverted evidence-quality pattern worth flagging to curators.

### Paralogs / ambiguity (3)

- **PP\_2916 ribC (Q88IT3, 209 aa, TrEMBL).** Second riboflavin synthase; InterPro confirms genuine two-domain riboflavin-synthase fold. **Genuine paralog, not an error.** Ectopic (not in the rib cluster). Which copy is physiologically primary is unresolved.
- **PP\_0516 ribAB-I (Q88QH7, 369 aa, TrEMBL) and PP\_3813 ribAB-II (Q88GB1, 373 aa, TrEMBL).** Both are two-domain proteins: an N-terminal DHBP-synthase RibB domain plus a C-terminal domain that InterPro assigns to the GTP-cyclohydrolase-II superfamily (fold), so the local metadata symbol "ribAB" implies bifunctionality. **Direct active-site inspection refutes this:** the reviewed monofunctional RibA (PP\_0522) retains the intact GTP-CHII catalytic core `SECLTGD...CDCGSQ` (the ECLTGD catalytic motif + the three-cysteine CDCG zinc cluster), whereas the C-terminal domains of PP\_0516 and PP\_3813 contain **neither** the zinc-cysteine cluster (no CxC[G/S]) **nor** the EC..GD catalytic motif, and align only trivially to RibA (Smith-Waterman windows of just 15–33 residues). Loss of the catalytic zinc-coordinating cysteines abolishes GTP cyclohydrolase II activity — the hallmark **RibBX** phenotype

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**Conclusion:** these are DHBP-synthase-active RibB/RibBX proteins with a catalytically dead C-domain. UniProt's "DHBP synthase only" (EC 4.1.99.12) recommended name is **correct**; EC 3.5.4.25 and the "ribAB" symbol should NOT be assigned. Step 1 rests solely on the standalone reviewed RibA (PP\_0522). (Strong sequence-based inference, not experimental proof.)

## Likely over-propagated / out-of-scope (5)

- **PP\_0236 ssuE (Q88R97, reviewed)** and **PP\_2764 msuE (Q88J85, reviewed)** — NAD(P)H:FMN reductases (EC 1.5.1.38) for sulfonate monooxygenase systems. Flavin *utilization*, not synthesis.
- **PP\_0548 ubiX (Q88QE6)** — flavin prenyltransferase making prFMN for UbiD decarboxylases (EC 2.5.1.129; primary bucket ppu00627)

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Distinct specialized cofactor.

- **PP\_1674 bluB (Q88MA0)** — 5,6-dimethylbenzimidazole synthase (EC 1.13.11.79); B12 lower-ligand biosynthesis, consumes FMNH<sub>2</sub>.
- **PP\_4919 nudF (Q88DA8)** — Nudix ADP-ribose pyrophosphatase (EC 3.6.1.13).
- **PP\_5231 had (Q88CF0)** — (S)-2-haloacid dehalogenase (EC 3.8.1.-); not flavin-related.

## 5. Gaps, ambiguities, and likely over-annotations

- **No true gaps.** Every de novo + activation step is encoded.
- **RibBX correction** (curation fix): do NOT add EC 3.5.4.25 to PP\_0516/PP\_3813 and do NOT use the "ribAB" symbol — their C-terminal GTP-CHII-fold domain lacks the catalytic zinc-cysteine cluster and is catalytically dead (RibBX). Rename toward *ribB/ribBX*. GTP cyclohydrolase II is a **single-copy** step carried by RibA (PP\_0522).
- **Over-annotation into ppu00740:** ssuE, msuE, ubiX, bluB, nudF, had are present only because they consume/produce flavins or share EC-class fragments. They inflate the apparent gene count (15) relative to the ~7–9 true module members.

- **Nomenclature ambiguity:** ribC (PP\_2916) vs ribE (PP\_0515) both = riboflavin synthase; distinct from the *E. coli* ribC=RibF convention.
- **Evidence-quality inversion:** core enzymes are TrEMBL; peripheral enzymes are Swiss-Prot.

## 6. Module and GO-curation recommendations

**Module step status:** - Steps 1–8: **covered**. - Step 1 (GTP-CHII): **covered as a single copy by PP\_0522**; PP\_0516/PP\_3813 do NOT contribute (RibBX, catalytically dead C-domain) — mark their EC 3.5.4.25 claim as **module\_needs\_revision / reject**. - Step 6 (riboflavin synthase): covered by PP\_0515; PP\_2916 = **covered (redundant paralog)**.

**Module membership edits:** - **Remove/downgrade** ssuE, msuE, ubiX, bluB, nudF, had from the de-novo-biosynthesis module (mark `not_expected_in_target_taxon`/out-of-scope for this module; they belong to sulfur assimilation, ubiquinone/aromatic decarboxylation, B12, and nucleotide/dehalogenase modules respectively). - **module\_needs\_revision:** generic module lists GTP-CHII only under "RibA/RibBA" — for KT2440 the canonical activity is the *standalone* RibA (PP\_0522); the RibBA copies are lineage-specific extras of uncertain catalytic status.

**GO / annotation requests:** - Consider GO annotation `GO:0003935` (GTP cyclohydrolase II) for PP\_0516/PP\_3813 only after active-site confirmation. - No new GO terms appear necessary; existing terms cover all activities.

## 7. Genes to promote to full `fetch-gene` review

1. **PP\_0516 (ribAB-I)** and **PP\_3813 (ribAB-II)** — sequence analysis indicates degenerate **RibBX** (dead C-terminal GTP-CHII domain, EC 3.5.4.25 absent); confirm experimentally and correct EC/gene symbol to *ribB/ribBX*.
2. **PP\_2916 (ribC)** — establish that it is a functional second riboflavin synthase and its physiological role vs operonic PP\_0515.
3. **PP\_0602 (ribF)** — TrEMBL; central bifunctional flavin-activation enzyme deserves reviewed-level curation.

4. (Lower priority) **PP\_0514 ribD** and **PP\_0515 ribE** — TrEMBL core genes worth Swiss-Prot-level review.

## 8. Key references

- Brutinel ED, Dean AM, Gralnick JA. *Description of a riboflavin biosynthetic gene variant prevalent in the phylum Proteobacteria*. J Bacteriol. 2013.

**P** 24097946

(ribBA→ribBX misannotation; ~40% of Proteobacterial ribBA lack GTP-CHII activity.)

- Fassbinder F, Kist M, Bereswill S. *Structural and functional analysis of the riboflavin synthesis genes... from Helicobacter pylori*. FEMS Microbiol Lett. 2000.

**P** 11024263

(Why a standalone ribA is retained alongside ribBA.)

- Serrano A, et al. *Key residues at the riboflavin kinase catalytic site of the bifunctional riboflavin kinase/FMN adenylyltransferase*. 2013.

**P** 22892871

(Single bifunctional RibF/FADS converts riboflavin→FMN→FAD.)

- Wang Z, Khusnutdinova AN, et al. *Biosynthesis and Activity of Prenylated FMN Cofactors*. 2018.

**P** 29551348

Khusnutdinova AN, et al. *Prenylated FMN: Biosynthesis, purification, and Fdc1 activation*. 2019.

**P** 31072498

(UbiX makes prFMN for UbiD — out of scope for FMN/FAD activation.)

- Yurgel SN, et al. *Sinorhizobium meliloti flavin secretion... bifunctional RibBA protein*. 2014.

**P** 24405035

(RibBA duplication / flavin secretion context.)

- UniProt/InterPro (proteome UP000000556) — accessions Q88QH9, Q88QH8, Q88QH7, Q88QH6, Q88QH1, Q88QG4, Q88Q93, Q88IT3, Q88GB1, Q88QE6, Q88R97, Q88J85, Q88MA0, Q88DA8, Q88CF0.
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## Evidence basis and limitations

All conclusions are **homology/architecture-based** (UniProt recommended names, InterPro domains, sequence lengths, operon context). **No KT2440-specific biochemical or genetic (knockout/complementation) data** were located for the rib genes; transfer from related bacteria (*E. coli*, *H. pylori*, *S. oneidensis*, *S. meliloti*) is strong for pathway logic but does not confirm the catalytic status of the KT2440 ribBA/ribBX and ribC paralogs. Definitive resolution requires: (i) active-site residue analysis or enzyme assays for PP\_0516/PP\_3813 GTP-CHII, (ii) complementation/expression tests for PP\_2916, and (iii) upgrading core TrEMBL entries to reviewed status.

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